

Towards Financial Quotient Education for Adolescents with LLM-Powered Assistants

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Abstract

Financial Quotient Education (FQE) is a crucial component of compulsory education for adolescents. However, existing approaches encounter systemic challenges, including the high cost of training professional instructors and the limited effectiveness of non-systematic curricula. Large Language Models (LLMs) have recently demonstrated remarkable potential in education, offering advantages such as low deployment costs and powerful capabilities, making scalable adolescent FQE a possibility. This paper introduces Financial Quotient Education Teaching Assistant (FQE-TA), the first LLM-Powered Assistant designed explicitly for adolescent FQE, comprising features: (1) a database of prompts covering the six core dimensions of the FQE that human experts critically designed and reviewed; (2) a suite of tailored prompts designed to make the outputs of the LLM appropriate for different age groups. Moreover, an experimental design routine is provided to support future empirical studies on FQE. This FQE-TA offers a promising basis for future research by facilitating low-cost deployment and specialising in FQE.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Human-Centered Computing, Empirical Studies in HCI, Financial Quotient Education, Large Language Model, AI4Education

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1 Introduction

Financial Literacy Education (FLE) is receiving widespread attention from the educational community as an essential component of modern comprehensive education systems. FLE refers to systematic educational activities designed to cultivate individuals' financial competencies and money management capabilities, encompassing the knowledge, skills, and behavioral norms necessary for personal financial management. As specialized educational activities for developing learners' financial literacy, FLE is particularly crucial for adolescent populations. Due to its foundational role in creating financial awareness, management concepts, and economic thinking, adolescent FLE has been progressively integrated into primary and secondary education curricula and has begun to establish systematic teaching methodologies and practical frameworks.

Existing research on adolescent FLE primarily adopts two teaching approaches [1, 2, 5, 8]. Traditional classroom instruction relies on teacher-led lectures and structured exercises, which can establish solid knowledge foundations but are constrained by faculty resources and class scheduling; technology-enhanced methods improve learning engagement through gamified applications and interactive tools, demonstrating promising educational outcomes while still facing challenges in technical support and effectiveness assessment. The limitations of these existing approaches provide opportunities for exploring new teaching modalities.

Current adolescent financial literacy education primarily relies on professionally trained teachers delivering instruction in classroom environments. However, this traditional teaching model faces significant challenges in implementation. First, financial literacy courses often suffer from dispersed class scheduling, low teaching frequency, and insufficient continuity, which severely undermines

the effectiveness of traditional centralized teaching in knowledge transmission due to their fragmented organizational structure. Second, faculty with specialized backgrounds in financial literacy education and extensive teaching experience are incredibly scarce, while the costs of training and hiring such professional teachers are considerably high. Schools are often reluctant to bear additional personnel costs for financial literacy education due to budget constraints, resulting in difficulties ensuring effective teaching quality. These practical issues underscore the need to explore new teaching models that are cost-effective, highly efficient, and adaptable to flexible instructional arrangements.

The limitations of traditional FLE have prompted the search for new solutions. Moreover, the rapid development of large language model technology has provided new possibilities for addressing the aforementioned educational challenges [7, 11]. Large language models possess distinctive characteristics, including low deployment costs, extensive knowledge coverage, and strong scalability, which have enabled them to achieve successful applications across multiple domains, such as industrial manufacturing [4], medical diagnosis, and scientific research [6, 9, 10, 12]. In the educational sector, with the advent of the digital education era, artificial intelligence technologies such as large language models are being deeply integrated with various educational elements, driving the proliferation of "human-AI collaborative" learning models and creating unprecedented technical conditions for achieving personalized and adaptive education.

Motivated by the above empirical evidence, this study represents the first systematic attempt to integrate large language models into the field of financial literacy education, aiming to provide an effective solution to the organizational difficulties and unstable teaching quality currently facing financial literacy education. This cross-disciplinary application confronts two critical technical challenges. The first challenge is the lack of specialized knowledge curation and prompt engineering resources tailored to the financial literacy education domain, as well as exceptionally professional prompt libraries suitable for large language model training and optimization. The second challenge is the absence of effective evaluation frameworks to assess the applicability of different large language models in financial literacy education scenarios, making it difficult to determine which model configurations are most suitable for the learning needs of students across various age groups.

This study introduces a comprehensive large language model solution, the Financial Quotient Education Teaching Assistant (FQE-TA), designed specifically for financial literacy education. Our main contributions include three key components: (1) We constructed a financial literacy education prompt database critically designed and reviewed by domain experts, encompassing six core dimensions of financial literacy education. (2) We developed customized prompting strategies tailored to different age groups, enabling outputs to better align with students' cognitive levels and learning needs. (3) We provide a comprehensive experimental design framework, establishing a methodological foundation for future empirical research in financial literacy education. This lightweight system achieves cost-effective deployment while establishing a solid foundation for continued research and application development in this domain.

2 Preliminaries

DeepSeek-V2-Lite [3] is a lightweight large language model based on the Mixture-of-Experts architecture, featuring a total of 16B parameters with only 2.4B parameters activated during inference. The model is trained from scratch using 5.7 trillion high-quality tokens and outperforms dense models and traditional MoE architectures of comparable scale across multiple English and Chinese benchmarks. The model can be deployed on a single 40GB GPU while supporting efficient fine-tuning in 8×80GB GPU environments, providing the research community with an accessible and high-performance research platform.

The technical implementation of DeepSeek-V2-Lite is based on Multi-head Latent Attention (MLA) and the DeepSeekMoE expert mixing system. Multi-head Latent Attention significantly reduces memory consumption through latent vector compression, reducing the traditional attention's $O(nd)$ key-value cache complexity to $O(d_c)$, where $d_c \ll nd$. The DeepSeekMoE architecture achieves efficient computation through sparse expert activation, with its core computation represented as:

$$y = \sum_{i=1}^K G(x)_i \cdot E_i(x) \quad (1)$$

where $G(x) = \text{softmax}(xW_g)$ is the routing function, $\{E_1, E_2, \dots, E_N\}$ represents N expert networks, K is the number of activated experts with $K \ll N$. This architecture enables the model to maintain a total parameter scale of $O(N)$ while achieving an inference complexity of $O(K)$, thereby realizing an effective balance between computational efficiency and model capacity.

To adapt to the edge computing and lightweight requirements of financial literacy education in actual campus environments, we developed a lightweight Financial Quotient Education Teaching Assistant (FQE-TA) based on the DeepSeek-V2-Lite model to address existing challenges. The system fully leverages the lightweight advantages of DeepSeek-V2-Lite, providing schools with an affordable and easily deployable intelligent teaching solution.

3 Six-Dimensional Financial Literacy Education

3.1 Motivation

Existing financial literacy education lacks specialized prompt libraries designed specifically for financial literacy education. To address the absence of specialized knowledge curation and prompt engineering for large language models in financial literacy education applications, we have constructed a comprehensive financial literacy education prompt system. This system aims to provide large language models with a structured framework of financial literacy knowledge, ensuring the professionalism, accuracy, and educational applicability of model outputs, thereby achieving standardized and intelligent financial literacy education.

3.2 Overall

Our financial literacy education prompt system encompasses six core dimensions: savings, risk, income, investment, consumption, and credit, with each dimension rigorously reviewed and validated by multiple experts in the field of financial literacy education. The

savings dimension covers savings awareness cultivation and savings method selection; the risk dimension includes risk identification, assessment, and management strategies; the income dimension involves understanding and planning income sources; the investment dimension covers basic investment knowledge and strategy formulation; the consumption dimension includes consumption concepts and rational consumption decision-making; the credit dimension involves credit concepts and debt management knowledge. These six dimensions constitute a comprehensive financial literacy education knowledge system, providing a framework for developing financial literacy among students of different age groups.

3.3 Method

Our financial literacy education prompt system encompasses six core dimensions: savings, risk, income, investment, consumption, and credit, with each dimension rigorously reviewed and validated by multiple experts in the field of financial literacy education. We provide 250 multi-turn dialogue pairs for savings, 259 multi-turn dialogue pairs for risk, 238 multi-turn dialogue pairs for income, 226 multi-turn dialogue pairs for investment, 271 multi-turn dialogue pairs for consumption, and 203 multi-turn dialogue pairs for credit, with each dimension containing diverse teaching content and rich educational scenarios.

- (1) The savings dimension encompasses savings awareness cultivation, savings method selection, and savings goal setting, guiding students to understand the importance and basic principles of saving, and helping them establish correct saving concepts and develop good saving habits.
- (2) The risk dimension encompasses risk identification, risk assessment, and risk management strategies, cultivating students' cognitive abilities to various financial risks. It teaches students how to identify potential risks and formulate corresponding countermeasures.
- (3) The income dimension involves understanding income sources, income planning, and income growth strategies, guiding students to form correct income concepts, understand the characteristics of different income sources, and cultivate career planning awareness.
- (4) The investment dimension encompasses basic investment knowledge, an understanding of investment tools, and the formulation of investment strategies, providing students with systematic education on investment and financial management, thereby helping them understand fundamental investment concepts and risk-return relationships.
- (5) The consumption dimension encompasses consumption concepts, decision-making, and rational consumption, enabling students to establish scientific principles of consumption, learn to plan consumption expenditures rationally, and avoid impulsive and excessive consumption.
- (6) The credit dimension involves credit concepts, lending principles, and debt management knowledge, cultivating students' correct credit awareness, understanding basic lending principles, learning to manage personal debt, and reasonably maintaining good credit records.

In particular, our method is specifically designed to accommodate both adult and children's versions, adapting to the learning

needs of different age groups. The children's version utilizes vivid and imaginative language expressions, conveying financial literacy concepts through storytelling scenarios and gamified elements, emphasizing intuitive experiences and engaging learning, making it suitable for elementary and middle school students. The adult version employs more rigorous academic language and in-depth analytical discussions, focusing on logical reasoning and practical applications, including more complex financial concepts and case analyses, and is suitable for high school students and adult learners. This age-adaptive design ensures the alignment between teaching content and learners' cognitive levels, improving the targeting and effectiveness of financial literacy education. We fine-tuned the DeepSeek-V2-Lite using the above prompts to obtain the final FQE-TA.

4 Experiments

In this section, we describe the design guidelines for our future experiments.

4.1 Experimental Design

4.1.1 Participants. This study encompasses three age groups: elementary school students (ages 9-12), high school students (ages 15-18), and university students (ages 18-22), with university students as the primary focus of research. The experiment involves different types of educational institutions, including higher education institutions, urban high schools, and rural schools.

4.1.2 Group Design. A five-group controlled experimental design is employed, with specific groupings as follows:

- (1) **Blank Group:** No financial literacy education intervention
- (2) **Control Group:** Human teachers only, without large language model assistance
- (3) **Experimental Group 1:** Human teachers combined with FQE-TA Professional Edition
- (4) **Experimental Group 2:** Human teachers combined with FQE-TA Children's Edition
- (5) **Experimental Group 3:** Human teachers combined with generic large language models (e.g., untuned ChatGPT or GLM-4)

4.1.3 Randomization and Sample Size. Students are randomly assigned to five experimental groups through stratified randomization, maintaining balanced group sizes while strictly controlling for teacher quality and curriculum content consistency. The blank group serves as a baseline control to evaluate pre-test and post-test changes in students without any financial literacy education intervention, with baseline balance verified through significance testing ($p > 0.05$).

Using a stratified sampling methodology, the total sample size consists of 1,200-1,500 students. Samples are evenly distributed across five groups, with 200-300 participants per group, maintaining balanced age distributions and school types within each group. The sample size was determined through G*Power software calculations, which were sufficient to detect the statistical effects of teaching methods and prompt libraries.

Table 1: Experimental Group Design Specifications, where Generic LLM, FQE-TA_{Pro}, and FQE-TA_{Child} denote Untuned Foundation Models, FQE-TA Professional Edition, and FQE-TA Child Edition, respectively.

Design Element	Blank	Control	Exp ₁	Exp ₂	Exp ₃
Intervention	No instruction	Human only	Human + LLM	Human + LLM	Human + LLM
LLM Type	–	–	FQE-TA _{Pro}	FQE-TA _{Child}	Untuned LLM
Curriculum	N/A	Identical content			
Instructors	N/A	Qualification-matched			
Sample Size	$N \geq 200$ per group				
Age Distribution	Elementary (9–12)				
	High school (15–18)				
	University (18/19–22)				
	Balanced per group				
Institutions	Higher education, urban & rural schools				
Randomization	Stratified (age & institution)				
Focus	Primary: University students				

4.2 Experimental Details

4.2.1 Experimental Procedure. The Ethics Review Committee has approved this research design and is subject to independent supervision. The experimental period spans 3 weeks (Week 1: pre-test, Week 2: teaching intervention, Week 3: post-test). An offline teaching model is employed, combined with personal computer equipment (Windows 11 operating system, RTX 3090 graphics card, stable network connection).

4.2.2 Sample Allocation. Total sample of 1200 participants (accounting for 10–20% attrition, 1500 actually recruited), distributed through stratified sampling by age group (elementary students 9–12 years: 70 people, high school students 15–18 years: 70 people, university students 18/19–22 years: 100 people), school type (higher education, urban high schools, rural schools: 80 people each), and geographical region (eastern, central, western). Students are randomly assigned to five experimental groups with 240 participants each, with baseline balance verified through t-tests and chi-square tests ($p > 0.05$).

4.2.3 Teacher Training. All human teachers receive 3 hours of standardized training, pass standardized assessments (score > 80 points), and strictly follow standardized lesson plans to ensure teaching quality consistency.

4.2.4 Teaching Arrangements. The teaching arrangements for each group are as follows:

- **Blank Group:** Participates in non-financial alternative courses (mathematics or English, based on national curriculum standards)
- **Control Group:** Receives traditional financial literacy education covering six dimensions (savings, budgeting, investment, loans, risk awareness, financial attitudes)
- **Experimental Groups 1-2:** Combine FQE-TA systems (professional/children’s editions, 30 minutes lecture + 30 minutes human-computer interaction)
- **Experimental Group 3:** Combines generic large language models (e.g., GLM-4) for instruction

4.2.5 Assessment Tools. Questionnaires employ layered design: elementary students use simplified PISA financial literacy assessment (15–20 minutes, 20–25 questions, focusing on savings and allowance management), high school students use People’s Bank of China financial literacy assessment (20–25 minutes, 25–30 questions, including interest and budgeting content), university students use enhanced financial literacy assessment (25–30 minutes, 30–35 questions, covering complex concepts like investment and loans). Questionnaires across age groups contain 10–15 overlapping core questions (30–40%), with the blank group additionally completing teaching effectiveness questionnaires (10–15 questions, 1–5 point scale).

4.2.6 Quality Control. Questionnaires validated through 100-person pilot testing for reliability (Cronbach’s $\alpha > 0.7$) and validity. Data cleaning process excludes samples with $> 25\%$ attrition rate. Statistical analysis employs ANCOVA and ANOVA, controlling for external influences (e.g., media exposure). Classroom instruction is supervised and recorded by observers to ensure teaching consistency, with equipment testing ensuring $< 5\%$ failure rate.

4.2.7 Hypothesis Verification. The experimental design supports scientific verification of four core hypotheses: C1 (incremental teaching effects of large language models), C2 (compatibility between age groups and prompt libraries), C3 (effectiveness comparison between FQE-TA and generic large language models), and C4 (universal effectiveness across different populations).

5 Conclusion

This paper introduces FQE-TA (Financial Quotient Education Teaching Assistant), the first LLM-powered assistant specifically designed for adolescent financial literacy education. The system addresses the high cost of instructor training and inadequate curriculum systems in traditional financial education through an expert-designed six-dimensional prompt database and age-appropriate adaptation mechanisms, providing a scalable and cost-effective solution for adolescent financial literacy education.

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